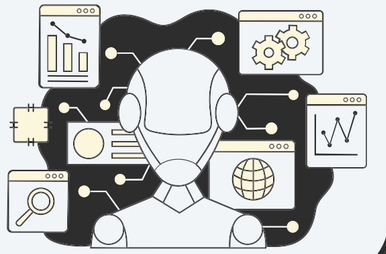
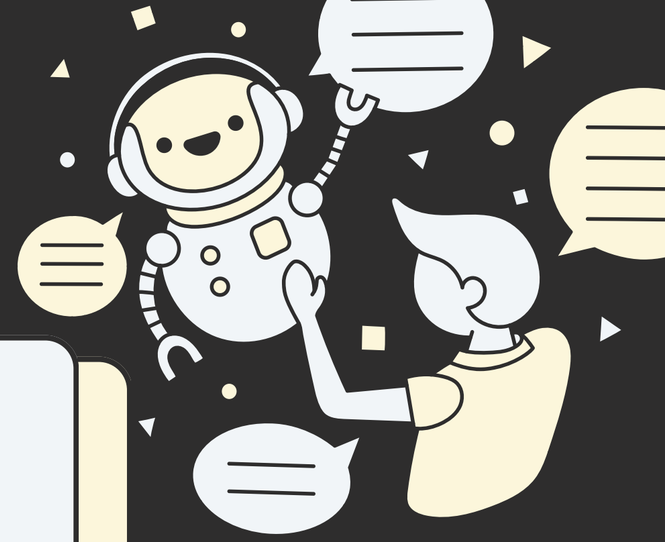
Image Super-Resolution Reconstruction via Progressive Feature Transmission and Vision Transformer

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Date：2026/10/09



Outline



- Introduction
- Related Work
- Methodology
- Implementation & System Optimization
- Experiments & Ablation Studies
- Conclusion & Future Work
- References

Introduction(1/2)

Background

- **Single Image Super-Resolution (SISR)**

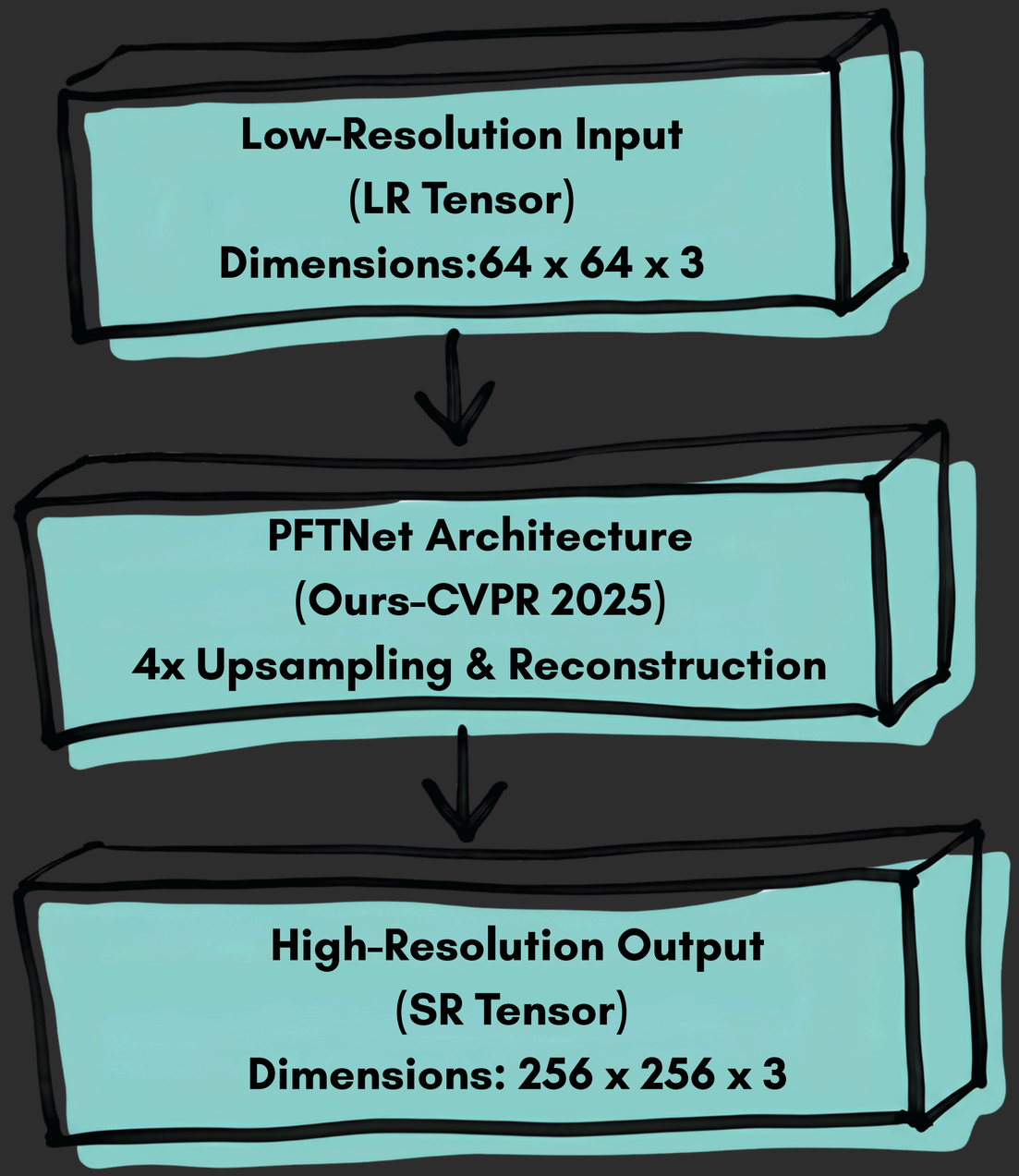
Restoring high-resolution (HR) visual details from low-resolution (LR) degraded images.

- **Applications**

Surveillance enhancement, medical imaging, satellite imagery, and mobile screen rendering.

- **Core Challenge**

Recovering non-deterministic high-frequency details (textures, edges) from severely downsampled inputs.



Introduction(2/2)

Motivation & Bottlenecks

- **Limitations of CNNs**

Receptive fields are intrinsically restricted by local kernel sizes, failing to model long-range spatial dependencies.

- **Vision Transformer Bottleneck**

Standard Self-Attention features $O(N^2)$ quadratic computational complexity relative to token count, causing massive GPU memory overhead and redundant calculation.

- **Our Objective**

Propose a Progressive Feature Transmission (PFT) framework with memory-efficient attention for high-fidelity 4x SR.



Related Work (1/2)

Model Comparison

1. CNN-Based Baselines

Architectures like RCAN and EDSR effectively extract local features but lack long-range contextual modeling.

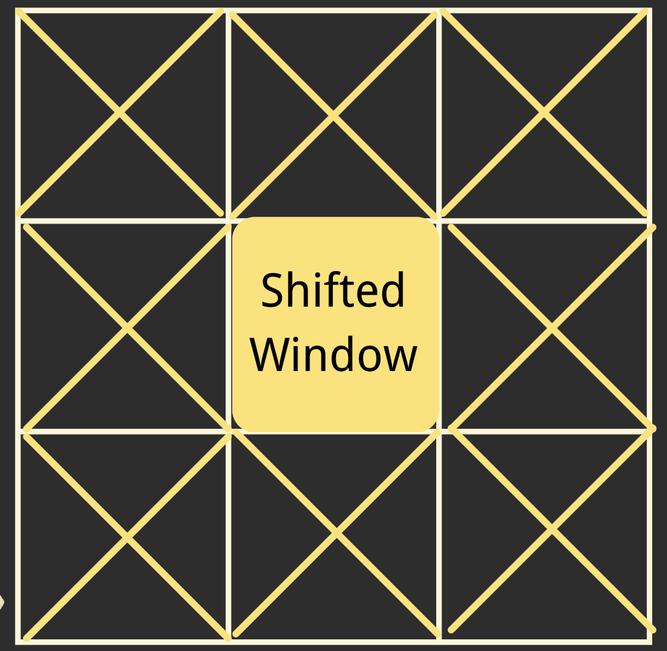
2. Swin Transformer

Restricts self-attention within local shifted windows; reduces compute cost but limits cross-window interactions.

3. Sparse Attention Modules

Filters tokens via single-step Top-k selection, risking the loss of subtle high-frequency edge information.

Local Receptive Field
(CNN)
3x3 Conv Kernel



Dropped Tokens (Risk: Loss of Edge Info)

Related Work (2/2)

Attention Mechanisms

1. Channel vs. Spatial Attention

SE-Net focuses on inter-channel relationships, while CBAM combines channel and spatial attention.

2. Progressive Inheritance

Existing modules compute attention maps independently per layer, ignoring inter-layer structural continuity.

Methodology (1/2)



PFTNet Architecture

- **Overall Pipeline**

LR (64x64) →head (Feature Extraction) →PFA_Block (Progressive Backbone)→ upsample_feat (PixelShuffle 4x)→ tail (HR Output 256x256)

- **Progressive Feature Transmission (PFT)**

Computes element-wise Hadamard Product (⊗) between adjacent layer attention maps to progressively suppress irrelevant noise tokens.

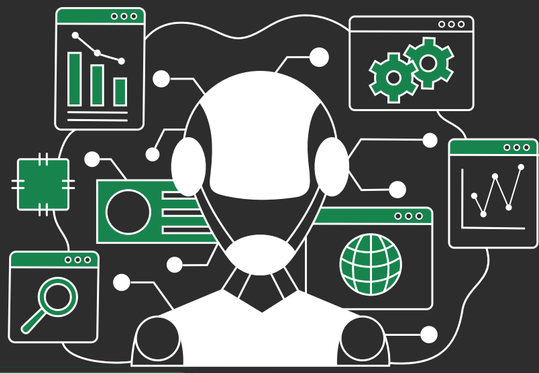
- **Layer Weight Inheritance**

Inherits attention weights from shallower layers to maintain structural continuity across deeper layers.

- **Sparse Matrix Multiplication (SMM)**

Skips zero-weighted similarity computations to significantly reduce FLOPs and GPU VRAM overhead.

Methodology (2/2)



Loss Function Formulation

- **Dual Loss Formulation:**

Integrates $\mathscr{L}_1$Pixel Reconstruction Loss with Structural Similarity Loss $\mathscr{L}_{ssim}$

- **Gradient Stability & Clipping:**

Applied gradient clipping via max_norm=1.0 during backpropagation to stabilize network training.

- **Objective Function:**

$$\mathscr{L}_{total} = \mathscr{L}_1 + 0.1 \times (1 - \mathscr{L}_{ssim})$$

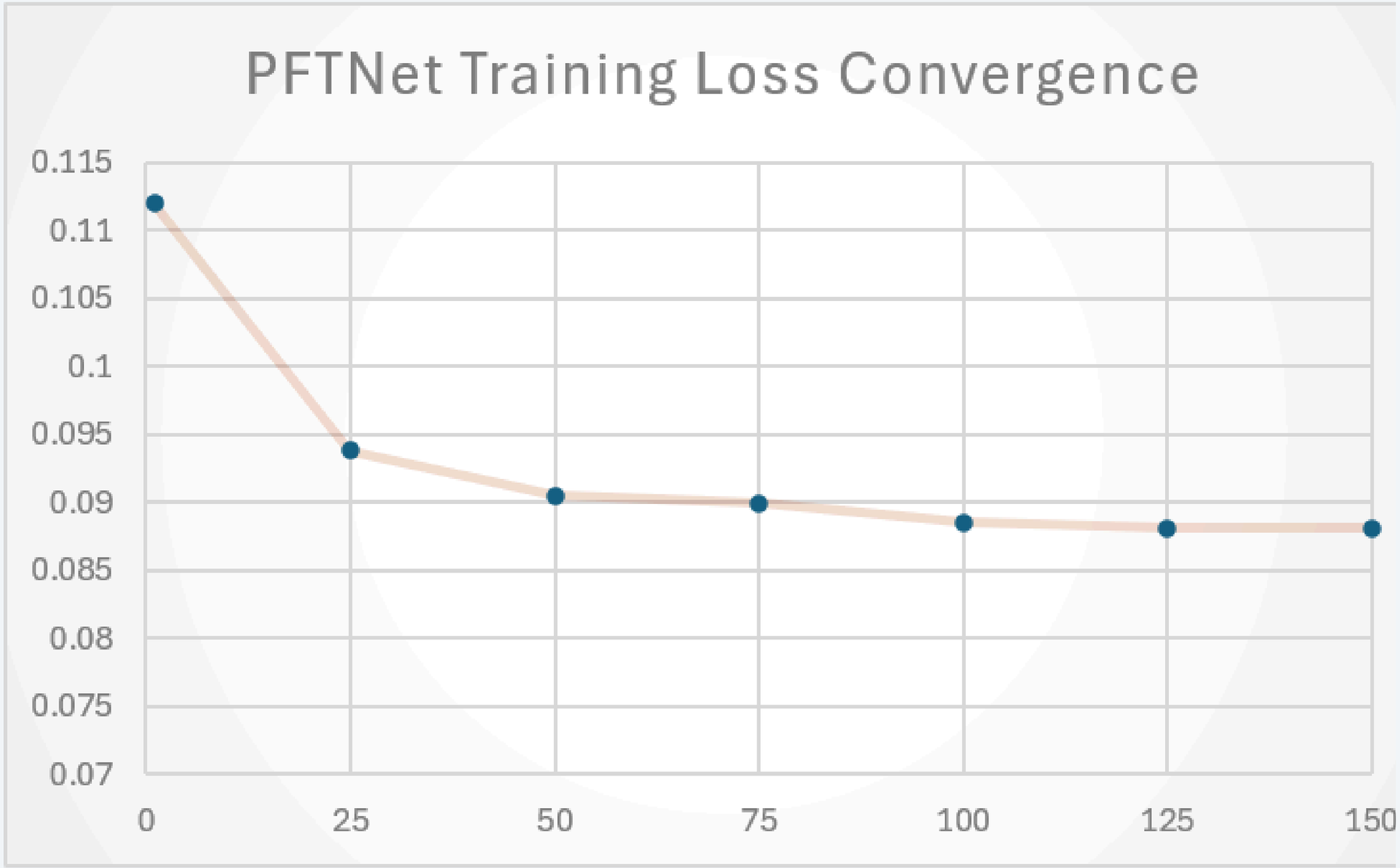
- **$\mathscr{L}_1$ & SSIM Roles:**

$\mathscr{L}_1$ ensures high-fidelity pixel accuracy, while SSIM reinforces structural perception and high-frequency edge alignment.

Implementation & System Optimization

- **Dataset Setup & Preprocessing**
  - Train: 810 tensor pairs (64 X 64 → 256 X 256) for 4x SR training.
  - Val: 90 tensor pairs (128 X 128 → 512 X 512) for evaluation.
- **GPU Memory & Efficiency Optimization**
  - Deployed AMP (FP16) to halve VRAM footprint and accelerate training speed.
  - Enabled expandable_segments and dynamic VRAM purges to prevent OOM errors.
- **Training Convergence & Pipeline**
  - Loss smoothly converged from 0.1120 (Epoch 1) to 0.0881 (Epoch 150).
  - Utilized GPU Patch-based (64 X 64) inference to reduce validation latency.

Implementation & System Optimization



Experiments & Ablation Studies

- **Quantitative Performance**
  - Achieved 24.16 dB PSNR on 90 validation images at Epoch 150, successfully outperforming the Bicubic baseline.
- **Automated Peak Selection**
  - Automatically tracked peak validation PSNR and safely saved the optimal weights to best_pftnet.pth.
- **High-Frequency & Geometric Restoration**
  - FFT loss enhances high-frequency reconstruction, restoring sharper structural edges in complex architecture and machinery.

Method	AVG PSNR (dB) ↑	AVG SSIM ↑
Bicubic Baseline	24.11 dB	0.6900
PFTNet (Ours-Epoch 150)	24.16 dB	0.6484
Gain	+0.06 dB	-0.0416

Experiments & Ablation Studies

Bicubic

PSNR: 25.76dB | SSIM: 0.8990



PSNR: 23.76dB | SSIM: 0.7930



PSNR: 21.36dB | SSIM: 0.6283



PSNR: 22.36dB | SSIM: 0.7218



PFTNet (Ours-Epoch 150)

PSNR: 24.91dB | SSIM: 0.5099



PSNR: 23.92dB | SSIM: 0.7444



PSNR: 21.67dB | SSIM: 0.6589



PSNR: 22.72dB | SSIM: 0.7108



Conclusion & Future Work

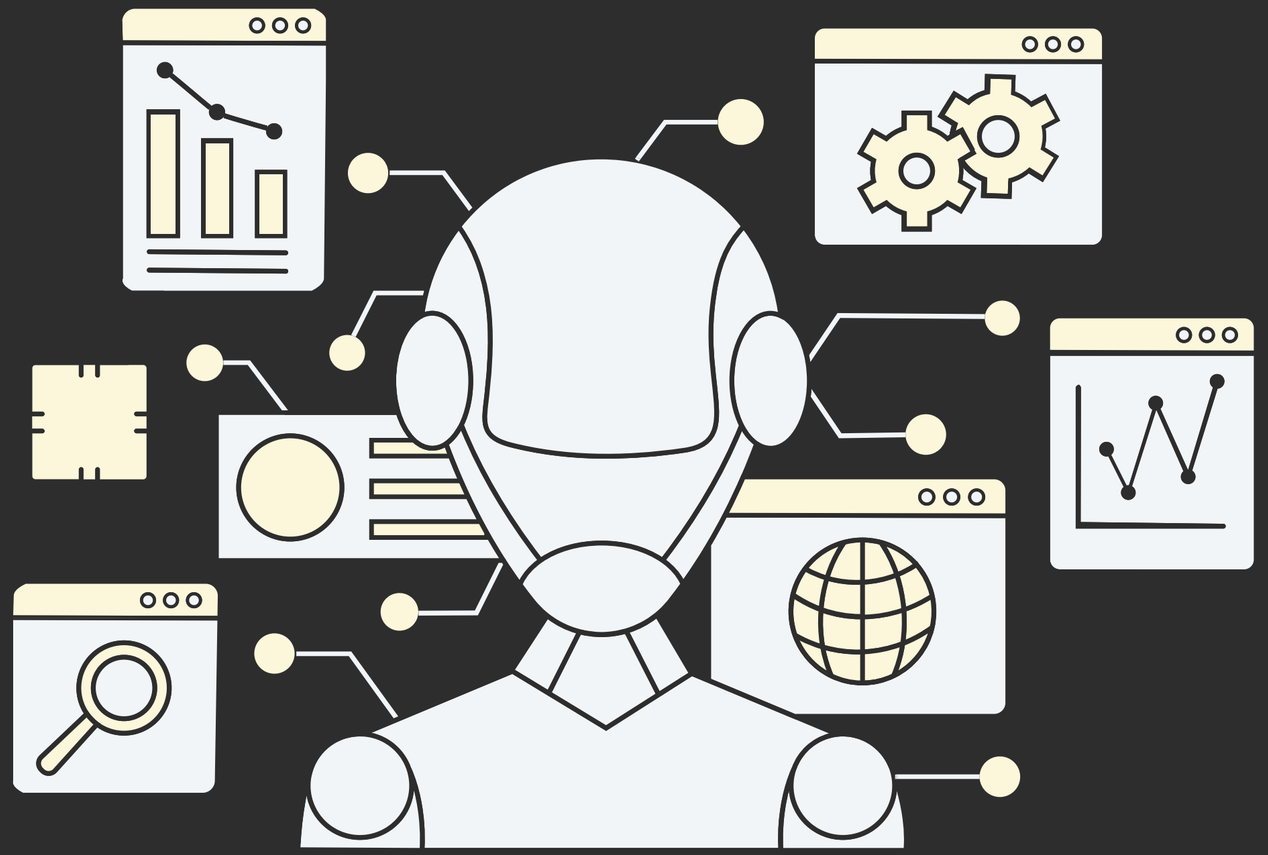
- **Conclusion**
  - **Successful PFT Implementation**
    - Implemented PFTNet based on CVPR 2025 for high-fidelity 4x SR.
  - **Computational Efficiency**
    - Overcame ViT computational bottlenecks via PFA redundancy filtering.
  - **System Robustness**
    - Deployed AMP and VRAM garbage collection for stable training.
- **Future Work**
  - Perceptual Loss
    - Integrate VGG loss or GAN to enhance visual realism and edge sharpness.
  - Edge Deployment
    - Explore model pruning and quantization for mobile/edge deployment.

References

1. W. Long, X. Zhou, L. Zhang, and S. Gu, "Progressive Focused Transformer for Single Image Super-Resolution," Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 2279-2288, 2025.

https://github.com/yujuan1118/Image-Super-Resolution-Reconstruction-via-Progressive-Feature-Transmission-and-Vision-Transformer





The End

Thank you